

```
In [1]: import pandas as pd
```

```
In [2]: data = pd.read_csv("trends.csv")
df = pd.DataFrame(data)
```

```
In [3]: df.head()
```

```
Out[3]:
```

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe

```
In [4]: df.tail()
```

```
Out[4]:
```

	location	year	category	rank	query
26950	Vietnam	2020	Là Gì?	1	Virus Corona là gì
26951	Vietnam	2020	Là Gì?	2	Miền thị thực là gì
26952	Vietnam	2020	Là Gì?	3	Đầu cắt moi là gì
26953	Vietnam	2020	Là Gì?	4	Bệnh bạch hầu là gì
26954	Vietnam	2020	Là Gì?	5	Đông Lào là gì

```
In [5]: df.shape
```

```
Out[5]: (26955, 5)
```

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 26955 entries, 0 to 26954
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   location    26955 non-null  object
1   year        26955 non-null  int64
2   category    26955 non-null  object
3   rank        26955 non-null  int64
4   query       26955 non-null  object
dtypes: int64(2), object(3)
memory usage: 1.0+ MB
```

```
In [16]: df.describe()
```

```
Out[16]:
```

	year	rank
count	26945.000000	26945.000000
mean	2015.241974	3.000000
std	3.564557	1.41424
min	2001.000000	1.00000
25%	2013.000000	2.00000
50%	2016.000000	3.00000
75%	2018.000000	4.00000
max	2020.000000	5.00000

```
In [8]: df.columns
```

```
Out[8]: Index(['location', 'year', 'category', 'rank', 'query'], dtype='object')
```

```
In [9]: df.dtypes
```

```
Out[9]: location    object
year            int64
category         object
rank            int64
query           object
dtype: object
```

```
In [10]: df.isnull().sum()
```

```
Out[10]: location    0
year              0
category          0
rank              0
query             0
dtype: int64
```

```
In [11]: df.nunique()
```

```
Out[11]: location      83
year                20
category           2450
rank                5
query             18431
dtype: int64
```

In this notebook, I did basic data analysis on the trends.csv dataset using Pandas.

I loaded the dataset using `pd.read_csv()` and checked the first and last five rows.

The dataset has 26,955 rows and 5 columns: location, year, category, rank, and query.

The data contains 2 numerical columns and 3 text columns. The years range from 2001 to 2020.

The rank values are between 1 and 5, with an average of 3.

There are 83 locations, 2,450 categories, and 18,431 different search queries.

There are no missing values in the dataset.

I also found 10 duplicate rows, which should be removed before further analysis

```
In [12]: df.duplicated().sum()
```

```
Out[12]: np.int64(10)
```

```
In [13]: duplicates = df[df.duplicated()]
print(duplicates)
```

	location	year	category	rank	\
20000	Kazakhstan	2018	Жылдың фильмі/Фильм года	1	
20001	Kazakhstan	2018	Жылдың фильмі/Фильм года	2	
20002	Kazakhstan	2018	Жылдың фильмі/Фильм года	3	
20003	Kazakhstan	2018	Жылдың фильмі/Фильм года	4	
20004	Kazakhstan	2018	Жылдың фильмі/Фильм года	5	
25400	Kenya	2020	Trending How To (Tech)	1	
25401	Kenya	2020	Trending How To (Tech)	2	
25402	Kenya	2020	Trending How To (Tech)	3	
25403	Kenya	2020	Trending How To (Tech)	4	
25404	Kenya	2020	Trending How To (Tech)	5	
			query		
20000			Веном		
20001			Мстители: Война бесконечности		
20002			Бизнес по-казахски в Америке		
20003			Монстры на каникулах 3		
20004			Дэдпул 2		
25400			How to fill Tpad online		
25401			How to create a website		
25402			How to connect a phone to TV		
25403			How to make my electric plate cooker work		
25404			How to take a screenshot on a laptop		

```
In [14]: df.drop_duplicates(inplace=True)
```

```
In [15]: df.duplicated().sum()
```

```
Out[15]: np.int64(0)
```

```
In [ ]:
```